

$$\| \chi^x(s) \|$$

$$\| \chi^x(s) \|_2^2 \leq D^* \int_0^s (s-r)^2 dr \leq \int_0^s \int_0^r G(s-r, y-z_1) \| \rho(u, r, z_1) \|_G f(z_1-z_2) dz_1 dz_2$$

$$G(s-r, y-z_2) \| \rho(u, r, z_2) \|_G$$

$$\int \| \chi^x(s) \|_2^2 \hat{\rho}(y) dy \leq C^* \int_0^s (s-r)^2 \int_0^r \int_0^r dz_1 dz_2$$

$$\times \| \rho(u, z_1) \|_G$$

$$\times \| \rho(u, z_2) \|_G$$

$$\times \left[f(z_1-z_2) \cdot G(2(s-r), z_1-z_2) \right]$$

$$\times \left[G(s-r, y+\frac{z_1-z_2}{2}) \hat{\rho}(y) dy \right]$$

$$\times \left[\hat{\rho}(s-r) \right]$$

e